FORMAL LANGUAGE

NATURAL VS FORMAL LANGUAGE

- Natural language – spoken language such as Bahasa Melayu, English, Spanish etc..
- Very complicated to specify all the rules of syntax for all natural languages.
- Very difficult to use natural language to communicate with the computer
- Formal language – model the natural language to enable human to communicate with the computer
- Has well-defined set of rules of syntax

GRAMMAR

- A grammar is a set of rules that describe how to form legal strings in the language.
- For English we have the following loose rule:

  sentence → **noun-phrase verb-phrase**

  which we interpret as saying:

  "A valid sentence consists of a noun-phrase followed by a verb-phrase"

- To complete the grammar, we then need to define **noun-phrase**, **verb-phrase** and so on, which are defined in the same way:

  noun-phrase → **article noun**

  verb-phrase → **verb adverb**

FORMAL GRAMMAR

- A formal *grammar* ***G*** is any compact, precise mathematical definition of a language ***L***.
- As opposed to just a raw listing of all of the language's legal sentences, or just examples of them, a grammar implies an algorithm that would generate all legal sentences of the language.
- Often, it takes the form of a set of recursive definitions.
- A popular way to specify a grammar recursively is to specify it as a *phrase-structure grammar (PSG)*

  $G = \{V, T, S, P\}$

Phrase Structure Grammar (PSG)

G = {V, T, S, P}

V = vocabulary

T = terminal

S = start symbol

P = production

Vocabulary

- A *vocabulary* (or *alphabet*) *V* is a finite, nonempty set of elements called *symbols*. A *word* (or *sentence*) over *V* is a string of finite length of elements of *V*.
- Contains terminals and non terminals elements.
- Terminals are represented in small letters whereas non terminals are represented by capital letters

Terminals

- Terminals are some of the elements of the vocabulary which cannot be replaced by other symbols.
- Represented in small letters

Production

The rules that specify when we can replace a string from V^*, the set of all strings of elements in the vocabulary, with another string are called the **productions of the grammar.**

Order of Production

Let G = ({S, A, B, a, b}, {a, b}, S, {S → AB, A → aAa | ε, B → Bb | ε})

- In each step the leftmost variable in the string is replaced
  - e.g. S ⇒ AB ⇒ aAaB ⇒ aaB ⇒ aaBb ⇒ aab
- In each step the rightmost variable in the string is replaced
  - e.g. S ⇒ AB ⇒ ABb ⇒ Ab ⇒ aAab ⇒ aab
- A grammar is ambiguous if there exist **two or more** distinct left-most (or right-most) derivations for a string *w* (i.e., two distinct derivation trees for *w*)
  - e.g. Grammar with productions {S → aSb | SS | ε}
  - S ⇒ aSb ⇒ aaSbb ⇒ aabb
  - S ⇒ SS ⇒ S ⇒ aSb ⇒ aaSbb ⇒ aabb

Phrase Structure Grammar (PSG)

Let G be the grammar with vocabulary $V = \{S, A, a, b\}$, set of terminals $T = \{a, b\}$, starting symbol S, and productions $P = \{S \to aA, S \to b, A \to aa\}$. What is $L(G)$, the language of this grammar?

$S \Rightarrow aA \Rightarrow aaa$

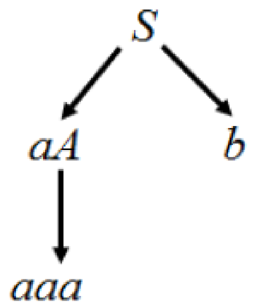
$S \Rightarrow b$

$L(G) = \{b, aaa\}$

Order of Production

- Another way to show derivation is by using a derivation tree

Let G be the grammar with vocabulary V = {S, A, a, b}, set of terminals $T = \{a, b\}$, starting symbol S, and productions $P = \{S \rightarrow aA, S \rightarrow b, A \rightarrow aa\}$. What is $L(G)$, the language of this grammar?



$L(G) = \{b, aaa\}$

Type 1 Context Sensitive

- A **type 1** grammar can have productions of the form $aA\beta \rightarrow a\gamma\beta$, where;
  - $A \in V$
  - $a, \beta \in (V + T)^*$
  - $\gamma \in (V + T)^+$
- $|aA\beta| <= |a\gamma\beta|$
- *Eg:*

  aAb → abbb

  aA → abb
- **context-sensitive** because *the derivation must be* surrounded by the strings *a* and *β*.
- A language generated by a type 1 grammar is called a **context-sensitive language**.

Type 2 Context Free

- A **type 2** grammar can have productions only of the form $A \rightarrow a$, where
  - A ∈ V
  - $A \in (V + T)^*$
- Type 2 grammars are called **context-free grammars** because a nonterminal symbol that is the left side of a production can be replaced in a string whenever it occurs, no matter what else is in the string.
- A language generated by a type 2 grammar is called a **context-free language**
- Eg:

  S → AB

  A → a

  B → b

Type 3 Regular

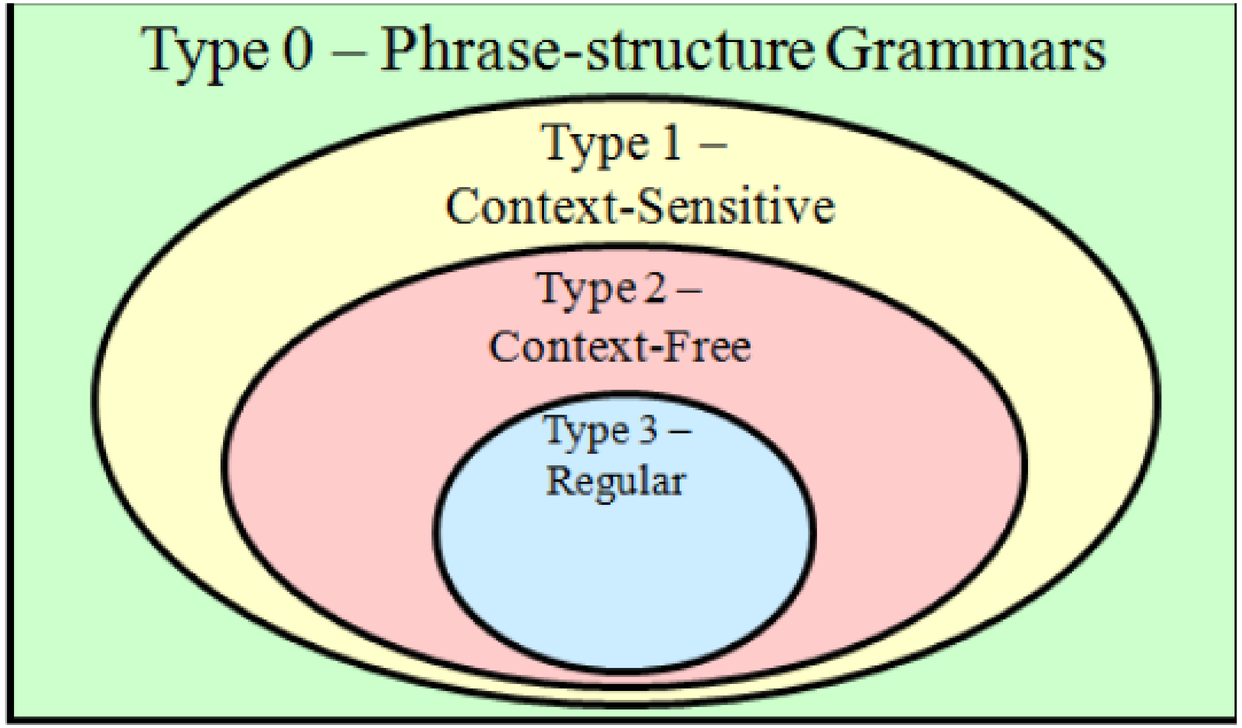
- A **type 3** grammar can have productions only of the form
  - $A \rightarrow aB$
  - $A \rightarrow a$
  - $A \rightarrow \varepsilon$

  for A, $B \in V$, $a \in \Sigma^*$, and ε the empty string
- Eg:

  $S \rightarrow aS|b$

  $S \rightarrow \varepsilon$

Types of Grammar



Let G be the grammar with $V = \{S, a, b, c\}$; $T = \{a, b, c\}$; starting symbol S; and productions $S \to abS$, $S \to bcS$, $S \to bbS$, $S \to a$, and $S \to cb$.

- Construct derivation trees for

  a) *bcbba*.

  b) *bbbcbba*.

  c) *bcabbbbbcb*.